Finstein

Frappe

Visitor Management

Modern visitor lifecycle for offices, factories, and campuses — from pre-visit invitation, through QR-based gate check-in and identity verification, all the way to hospitality, conference rooms, contact tracing, and audit reports. Built on Frappe & ERPNext.

ERPNext 16 · Frappe 16 · license MIT

Main features

Visitor Lifecycle (invitation → approval → gate → checkout): One Visitor Pass record drives the entire journey. Walk-in passes from reception or portal pre-registration via Visitor Invitation. Visitor type drives the approval workflow lane automatically.

Gate Security & Identity Verification: QR scan at the gate auto-loads the pass. A live photo is captured at the gate, **Matches ID Proof** and **Matches Pass Photo** must be ticked, the system screens against the active **Visitor Blacklist** (by ID number or name+type), issues a badge once items are verified, and records the gate event as an immutable **Security Log** row that can't be edited after save.

Hospitality Management: When a Visitor Pass is approved with any hospitality flag (cab, hotel, factory tour, buggy, greeting, meal), the system auto-creates a **Hospitality Request**. Meal slots are computed against the visit window; cab pick-up/drop times auto-populate from the pass.

Conference Room Booking: If a room is selected on the pass, a **Conference Room Booking** is auto-created. Meeting time is clamped to the room's operating hours and seating capacity is validated.

VIP Protocol: Dedicated **Approved VIP Queue** quick-action at the gate, email notification on pass approval, **protocol notes** carried from the pass through to the gate, pre-allocated meeting room, two-step approval.

Contact Tracing & Audit Trail: Every gate event opens or closes a **Contact Trace Record** for the visitor (visited area, time in, time out, exposure risk). Every lifecycle event is logged in the immutable **Visitor Event Log**.

Before you install — two things a buyer must know

These are the two decisions this app makes on your behalf. Read them before running `bench install-app`, not after.

1. Installing this app switches on site-wide guest file uploads

The visitor pre-registration form asks a not-logged-in visitor for a scan of their ID and a photo of themselves, and **both fields are mandatory**. Frappe refuses any upload from a not-logged-in caller unless the System Setting **Allow Guests to Upload Files** is on. So `bench install-app finstein_visitor_management` switches that setting on — and it is **site-wide**, affecting every app on the site, not only this one.

That setting is dangerous on its own. Frappe's upload handler treats it as blanket authority for a guest: it skips the write-permission check entirely and takes the target record from the request body. Left unfenced, anyone on the internet could attach a file to any record on your site.

What fences it: a `File` `before_insert` hook — `finstein_visitor_management/visitor_management/portal_upload.py`, wired at `hooks.py` `doc_events` — inspects every file a not-logged-in caller creates. An upload from the visitor pre-registration form must be unattached or aimed at **Visitor Pass → ID Proof Scan / Visitor Photo**; it must be a real JPG, PNG or PDF (checked by reading the file's own bytes, not trusting its name), be under 5 MB, stay within 30 uploads per hour per IP address, and carry the random key the form page made when it loaded — only the submission holding that key can claim the file, so two visitors on a shared kiosk cannot take each other's documents. Every such file is forced private, so it cannot be read by guessing a URL. Logged-in users of any app are not touched by this hook at all.

Other apps' public forms get exactly what they would get without this app:

- If **Allow Guests to Upload Files** was already on before you installed this app, guest uploads from every other page keep working as before.
- If this app is what switched it on, guest uploads from other pages are refused — the same as Frappe does with the setting off. To let another app's public form (a careers page taking CVs, say) accept uploads, list its path in **VMS Settings → Other Pages That May**

Accept Guest Uploads, one per line (for example `/job_application`).

What you must still do yourself:

- The setting is turned on **once, at install**. If you switch it off later, `bench migrate` will not switch it back on — it prints a note saying the portal cannot accept uploads.
- **Uninstalling the app switches it back off** — but only if this app is what switched it on. If it was already on before, it is left as it is and the uninstall says so.
- If you do not intend to use the guest pre-registration portal, switch the setting off after install. The rest of the app — walk-in passes, the gate, hospitality, rooms — works without it.

2. The app stores visitor identity data, and ID numbers are stored unencrypted

This section is written for a compliance officer. It describes what is on disk, not what is intended.

What is stored	Where	How
Visitor name, mobile number, email address	Visitor Pass, Visitor Invitation	Plain columns. The mobile number is also kept in a normalised digits-only form for matching.
Photograph of the visitor (uploaded at pre-registration, and captured live at the gate)	File attachments on Visitor Pass and Security Log	Private files — not readable by URL guessing.
Government ID **type and number** — PAN, Aadhaar, Passport, Driving Licence	Visitor Pass → `id_proof_type`, `id_proof_number`	**Plain text in an ordinary database column. Not encrypted.**
Scan or photo of the ID document	File attachment on Visitor Pass	Private file.
Movement history — every gate event, area entered and left, exposure risk	Security Log, Contact Trace Record, Visitor Event Log	Append-only in practice. A saved Security Log refuses further edits from everyone except System Manager, who may correct one; Contact Trace Record and Visitor Event Log hold no write permission for any role other than System Manager.

ID numbers are not encrypted. They are stored the same way any other text field is. Anyone who can read the database, restore a database backup, or holds the System Manager role can read every visitor's ID number in the clear. Frappe's field-level encryption is not applied to them, because the app matches on the stored value — the blacklist and duplicate checks look the number up directly, and an encrypted column cannot be searched that way.

Access controls that do apply:

- Attached photographs and ID scans are stored as **private files**, so a URL alone is not enough to fetch one — the viewer must be logged in and permitted.
- Security staff can read a Visitor Pass but cannot edit one; the gate acts through the Security Log, and a saved Security Log refuses any further edit unless the user is a System Manager.
- Permissions the app grants itself at migrate are granted one privilege at a time, and **read no longer silently brings `export` with it** (changed in 2.0.0). It is asserted once and then belongs to your Role Permission Manager — the app will not re-open a permission an administrator has tightened.
- On other apps' records — **Employee, Job Applicant, Supplier, Maintenance Visit** — the app grants its roles nothing: link fields search through a dedicated endpoint, and the few fields a pass copies from the picked record come from one that returns only those fields. (Earlier versions granted read, which on Employee is the whole HR record — bank account, salary, PAN, date of birth — or select. Migrating removes those rows once: only rows with exactly the rights the app wrote, for the roles it wrote them for. Rows an administrator has since changed, and roles HRMS or ERPNext grant themselves, are not touched.) Note that HRMS's own **Employee** role reads Employee records: give each employee user a User Permission to their own Employee (HRMS does this when "Create User Permission" is ticked on the Employee) or every employee can read every colleague's record.

Who can export identity data today. Be aware that the shipped DocType permissions do give `export` on Visitor Pass — which includes the ID number column — to **CEO, HOD, HR Manager, Sales Manager, Facility Manager, Hospitality Manager, Host Employee, Security and System Manager**. If your policy is that only a named data owner may download visitor identity data, remove `export` from the rest in **Role Permission Manager** after install. The app will not put it back.

Retention. The app ships a retention purge, and it is **off by default**. Nothing is ever anonymised or deleted until an administrator ticks **VMS Settings → Data Retention & Purge → Enable Data Retention Purge** and confirms a retention period (**Retention Period (days)**, which arrives pre-filled with 365 but does nothing while the switch is off). Leave it off until your compliance function has settled on a period.

Once it is on, a job runs nightly at 03:00 and works like this:

What it anonymises	The fields that identify a person — name, mobile number, email, government ID number, vehicle number — plus the ID-scan and photo files, on the Visitor Pass and on the Security Log rows for the same visit.
What it keeps	The visit itself: dates, host, gate, badge, workflow history and item verification. So "how many contractor visits last quarter" and the gate's own audit trail still answer correctly after the visitor's PAN or Aadhaar number is gone.
What it will not touch	Any visit that is not over. Only a pass that is Checked-Out, Rejected, Cancelled or flagged No-Show — and an invitation that is Submitted, Expired or Cancelled — is ever a candidate, and only once it is also older than your retention period.
What it never touches, at any age	**Visitor Blacklist.** A blacklist entry exists precisely to identify someone durably; anonymising it would quietly switch off blacklist matching for that person. The job does not go near that doctype.

It anonymises rather than deletes, which is what lets the statistics and the audit trail survive the purge. Purged fields are marked `[Purged – data retention]`. The job is batched and idempotent — a row it has already purged has nothing left to match, so re-running it is safe.

Still not implemented. Retention is a schedule, not a request queue. The app does **not** provide consent capture, a subject-access export, or a right-to-erasure workflow for one named individual — honouring "delete everything you hold on me, today" still means an administrator finding and clearing that person's records by hand. If you operate under GDPR, India's DPDP Act or a similar regime, treat those three as controls you must add around the app.

Installation

Prerequisites

Frappe bench	on the `version-16` branch
ERPNext v16	required — declared in `required_apps`, bench refuses to install without it
HRMS v16	required — the Employee doctype is used for hosts and approvers
Python	3.14 or newer
MariaDB	11.8 or newer, InnoDB
Node	24 (for `bench build`)

Install ERPNext and HRMS on the site first if they are not there already:

```
bench --site <your-site> install-app erpnext
bench --site <your-site> install-app hrms
```

Install

```
cd ~/frappe-bench
bench get-app --branch version-16 https://github.com/finstein-erpnext/Finstein_Visitor_Management
bench --site <your-site> install-app finstein_visitor_management
bench --site <your-site> migrate
bench build
bench restart
```

The repository is called `Finstein_Visitor_Management`, but bench reads the package name from `pyproject.toml` and puts it in `apps/finstein_visitor_management` — which is why `install-app` takes `finstein_visitor_management`, not the repository name.

Upgrade later

```
cd ~/frappe-bench/apps/finstein_visitor_management
git pull
cd ~/frappe-bench
bench --site <your-site> migrate
```

Migrations are idempotent — `bench migrate` is safe to run any number of times. All setup (roles, gates, visitor types, ID proof types, settings and the approval workflows) is applied on migrate, and re-applying it does not overwrite a choice you have since made in the Role Permission Manager.

Uninstall

```
bench --site your-site uninstall-app finstein_visitor_management
```

Besides the app's own DocTypes and records, the uninstall removes what the app added elsewhere: the Job Applicant interview fields, its three workflows and notifications, its permission rows on Employee / Supplier / Job Applicant / Maintenance Visit / Page (restoring those DocTypes' standard permissions when nothing else was customised), the roles it created, its scheduled jobs, and every visitor's ID scan, photo and visa copy on disk. It switches **Allow Guests to Upload Files** back off if the app switched it on. Roles or settings that existed before the app are left alone. `--dry-run` changes nothing.

Check it worked

Open **`/app/visitor-management`** — the Visitor Management workspace. On Frappe v16 the desk is served at `/desk`, and `/app` redirects there. The app also appears as a tile on the `/apps` screen.

Running behind a proxy

Visitors open their invitation link from outside the office, so two site settings matter in production.

`host_name` — the address invitation links are built on. Without it the link uses the address the host happened to send from, which on an office LAN can be an IP the visitor's phone cannot reach:

```
bench --site <your-site> set-config host_name "https://visitors.example.com"
```

`trusted_proxy_ips` — which proxies may tell the app the visitor's address. The public pre-registration form limits how often one visitor may submit and upload files. An invited visitor is counted against their own invitation, so nobody else's traffic can use up their allowance. A walk-in visitor (only when **Allow Pre-Registration Without Invitation** is on) is counted against their address, taken from `X-Forwarded-For` only when the request came from a trusted proxy. A proxy on the same machine (bench's own nginx) is trusted automatically. A proxy or load balancer on another machine must be listed. If it is not, every walk-in seems to come from the proxy's address, and they all share one allowance:

```
bench --site <your-site> set-config --parse trusted_proxy_ips '["10.0.0.4"]'
```

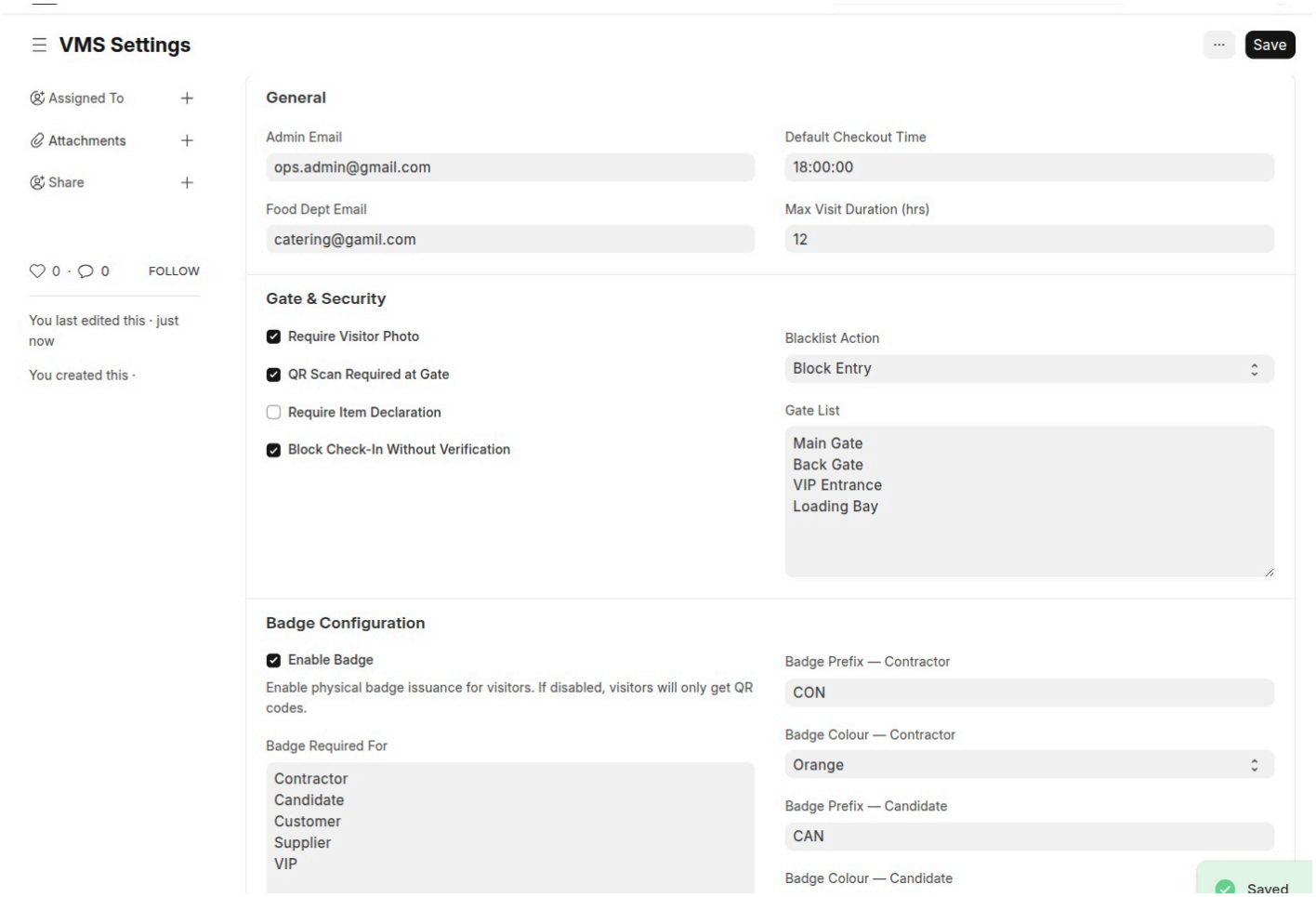
The proxy must **overwrite** `X-Forwarded-For` with the client's address, not append to it. Bench's nginx template already does this (`proxy_set_header X-Forwarded-For $remote_addr;`). With `$proxy_add_x_forwarded_for`, a visitor could choose their own address and get past the limit.

This README is the authoritative installation reference for v16. `docs/README.pdf` is a printable copy generated from it; regenerate it whenever this file changes.

Setup and Use

All system-wide configuration lives in a single doctype: **VMS Settings**. Open the desk Awesome Bar and type "VMS Settings".

Configure VMS Settings



VMS Settings — General, Gate & Security, and Badge Configuration sections

Field	What to put
Enable Badge	✅ to enable badge generation per visitor
Badge Required For	Comma-separated visitor types that get badges (e.g. `Contractor,Customer,Supplier`)
Badge Prefix — Contractor / Candidate / Customer / Supplier / VIP	Defaults: `CON / CAN / CUS / SUP / VIP`
Require Item Declaration	✅ to block pass submission until at least one item row is added
Auto-Cancel Pending Passes (Hours)	After how many hours a `Draft` pass auto-cancels (`0` to disable)
Default Gate	Gate name to fall back to when no auto-assignment rule matches

Click **Save**. The app is now configured.

Configure Workflows

Three approval workflows (Visitor Pass, Hospitality Request, Conference Room Booking) are pre-loaded as fixtures and activated automatically during app migration — no setup needed.

Roles — who does what

The app ships **11 roles**. Assign them alongside Frappe's standard **Employee** role: the everyday actions — raising a visitor pass, requesting hospitality, booking a room — are granted to `Employee`, and the roles below add the approval and operational rights on top.

Role	Who it is for	What the role can do
CEO	Executive sign-off	Reads, edits and approves Visitor Passes. Ships as the **second** approver on the VIP lane. Reads Hospitality Requests. Cannot create or delete either.
Facility Manager	Owns the meeting rooms	Full control of **Conference Room** and **Conference Room Booking** — create, edit, delete, and approve, reject, cancel or amend a booking. Reads Visitor Pass for context.
Factory Tour Coordinator	Runs plant and site tours	Reads **Hospitality Request**, and receives the 07:00 daily hospitality digest so the day's tours are known in advance.
Front Office Executive	Reception desk	Raises walk-in passes (through `Employee`) and can pick an existing **Supplier**, **Maintenance Visit** or **Job Applicant** on the matching pass layout. Receives the daily hospitality digest.
Greeting Staff	Meets visitors on arrival	Reads **Hospitality Request** and receives the daily digest — who is arriving, and what greeting was requested.
HOD	Department head	Reads, edits and approves Visitor Passes. Ships as the **first** approver on the VIP lane. Reads Hospitality Requests.
Host Employee	The person being visited	Creates, edits and submits **Hospitality Requests** for their visitors, and reads the Visitor Passes they host. Can pick a Supplier, Maintenance Visit or Job Applicant on a pass.
Hospitality Manager	Owns cabs, hotels, meals and tours	Full control of **Hospitality Request** — create, edit, delete, approve, reject, cancel and amend. Reads Visitor Pass.
Hospitality User	Hospitality team member	Reads **Hospitality Request** and receives the daily digest. An execution role, not an approver.
Security	The gate	Creates and edits **Security Logs** — check-in, check-out, gate transfer, alert, badge collected. Reads Visitor Pass, Visitor Gate, ID Proof Type and VMS Settings. **Read-only on Visitor Pass on purpose:** the gate acts through the Security Log, never by editing the pass.
Transport Coordinator	Cabs and drivers	Reads **Hospitality Request** and receives the daily digest with the day's pick-ups and drops.

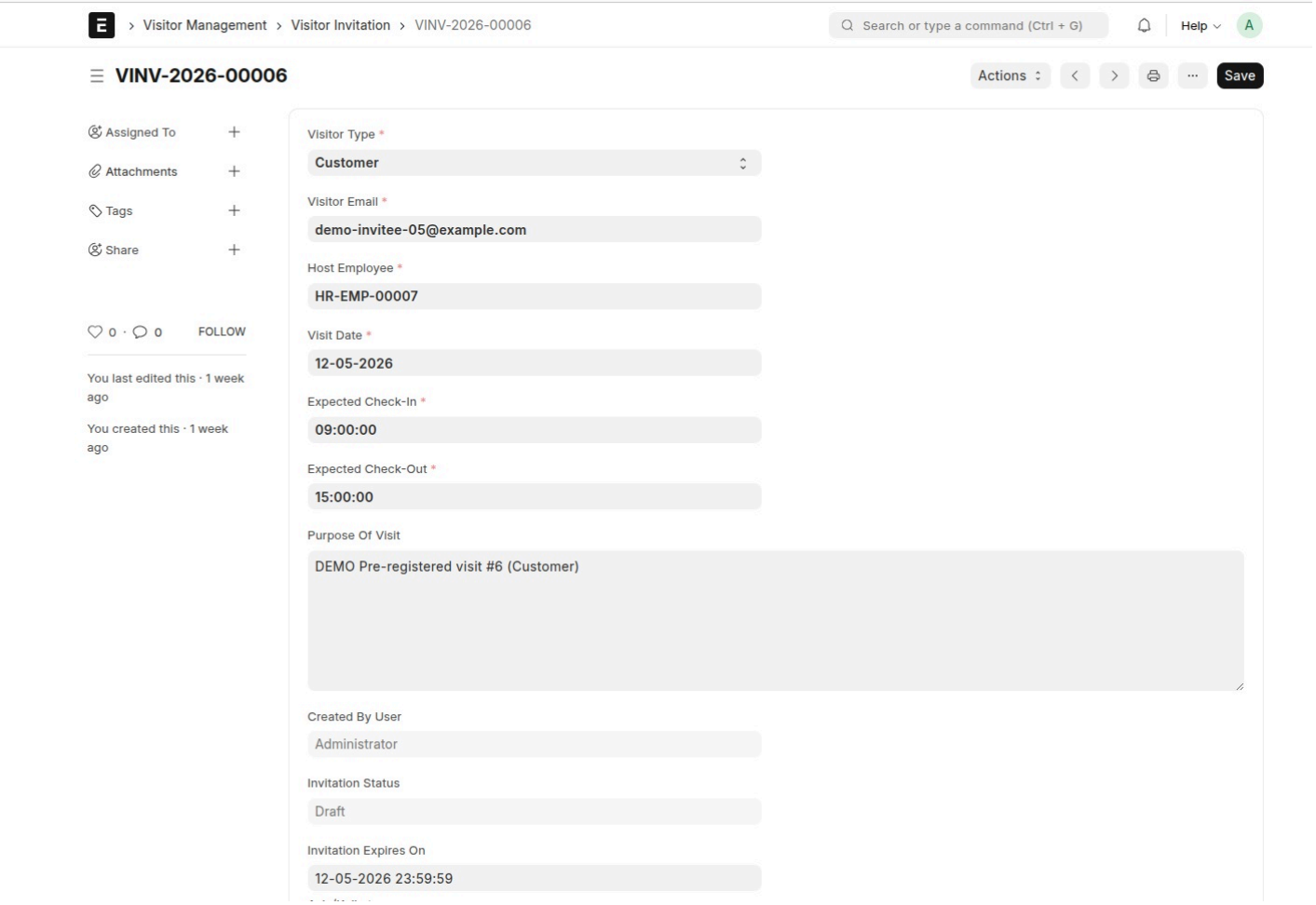
Approval routing is data, not code. Each **Visitor Type** names the role that approves it, and the Visitor Pass workflow is regenerated to match whenever you change one. The shipped defaults are:

Visitor Type	Approver	Second approver
VIP	HOD	CEO
Candidate	HR Manager	—
Customer	Sales Manager	—
Contractor	System Manager	—
Supplier	System Manager	—

Point a Visitor Type at a different role and run `bench migrate`; the workflow lane, the approval email and the Cancel permission all follow it.

Two standard ERPNext roles are used as shipped approvers and are **not** created by this app: **HR Manager** (Candidate lane) and **Sales Manager** (Customer lane).

Pre-Visit Invitation — invite a visitor in advance



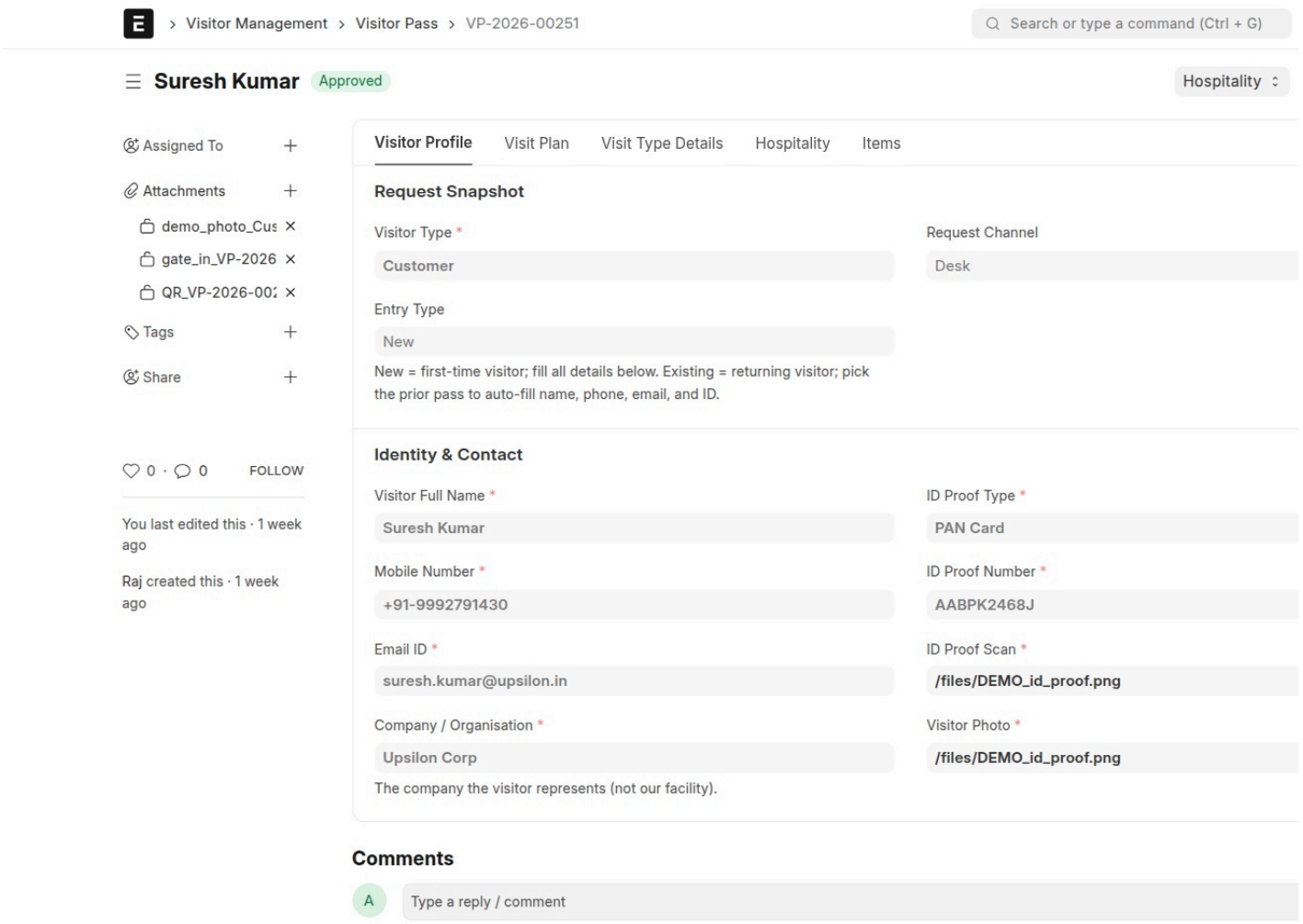
Visitor Invitation — pre-registration link generated for the visitor

1. Open the desk Awesome Bar and type "Visitor Invitation".
2. Click **+ Add Visitor Invitation**.
3. Fill: **Visitor Name**, **Email**, **Visit Date**, **Purpose**, **Person to Visit**.
4. Click **Save**. The system emails the visitor a token-based portal link.
5. The visitor clicks the link, fills the pre-registration form (mobile, ID type, ID number, photo, ID proof scan).
6. A **Visitor Pass** is auto-created in Draft state and routed to its approval lane based on visitor type.
7. On **Approve** → pass becomes Approved + Submitted, QR code is generated, notification is sent, hospitality request + room booking auto-create if flags are set.

End-to-end takes ~1 minute for the host, ~3 minutes for the visitor.

The link works once. It stops working when the visitor submits the form, when the invitation expires, or when it is cancelled. The invitation shows **Opened** when the visitor first uses the form. A mail scanner that only fetches the link does not count. One user can send up to 30 invitation emails an hour.

Walk-in Visitor — visitor is at reception

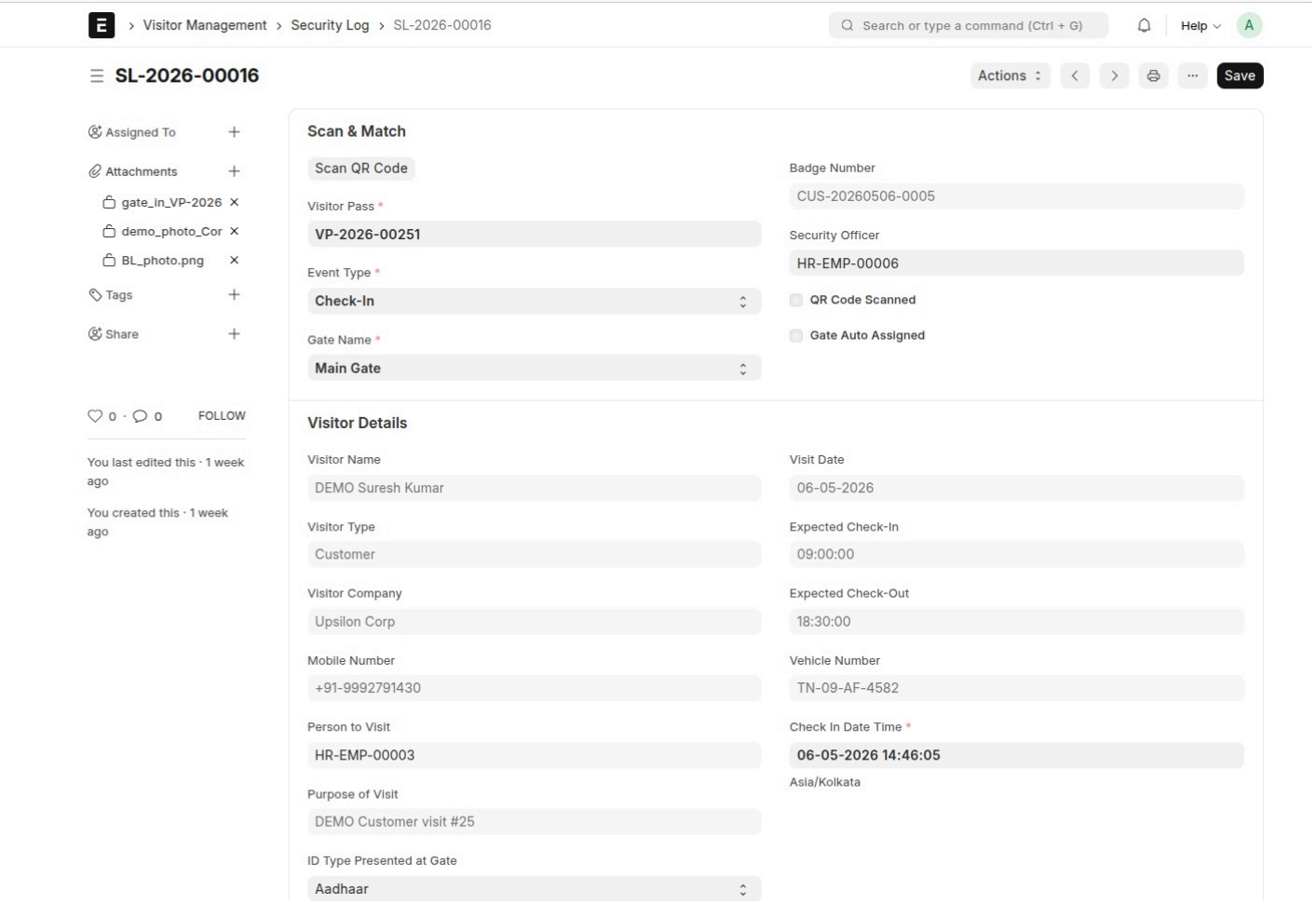


Visitor Pass — Approved state with QR code and host details

1. Open `/app/visitor-pass` and click **+ Add Visitor Pass**.
2. Fill: **Visitor Name**, **Mobile**, **ID Proof Type + Number**, **Visitor Type**, **Person to Visit**, **Visit Date**.
3. Tick hospitality flags if applicable (Meal Required, Cab Required, Hotel Required, etc.).
4. Click **Save**. Pass goes to Draft, then **Submit** to start the approval workflow.
5. On **Approve** → QR code generated, notification sent, hospitality + room booking auto-created.

End-to-end takes ~5 minutes including approval.

Gate Check-In — visitor arrives



Security Log Check-In — gate photo, identity match, item verification

1. Visitor presents QR code at the gate.
2. Open `/app/security-log` and click **Scan QR Code**.
3. The pass auto-fills — visitor photo, ID details, purpose, host, hospitality flags, items declared.
4. Click **Capture Gate Photo** → live photo is uploaded and attached.
5. Tick **Matches ID Proof** and **Matches Pass Photo** (both are mandatory).
6. Tick each declared item in the **Items Verification** grid as items are checked.
7. Click **Save**. Visitor Pass status flips to **Checked-In**, badge is issued, a notification email is sent, a Contact Trace Record opens.

The Security Log is **locked after save** — gate events are immutable audit records.

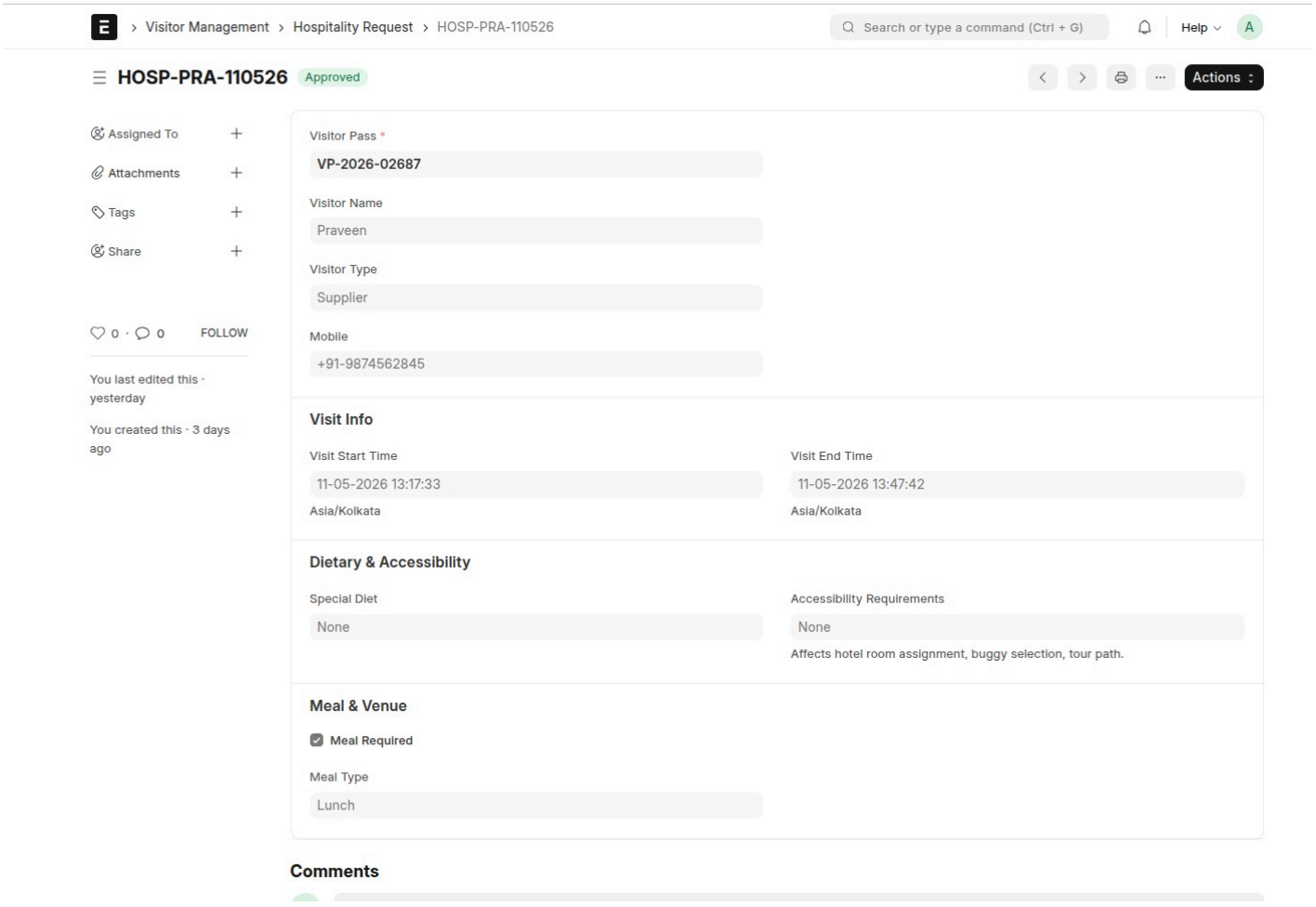
The system **re-checks the blacklist at the gate** even after pass approval — blacklisting may have happened in between. A blacklisted visitor is refused entry with a security alert email.

Gate Check-Out — visitor leaves

1. Visitor presents QR code at exit.
2. Create a new Security Log with **Event Type = Check-Out**.
3. Live photo + match confirmations (ID + pass photo) are mandatory — same standard as check-in.
4. Click **Save**. Visitor Pass flips to **Checked-Out**, the open Contact Trace Record closes, the badge is marked returned.

Visitors who get blacklisted while still on premises are **allowed to check out** (the goal is to keep them out, not trap them in).

Hospitality flow

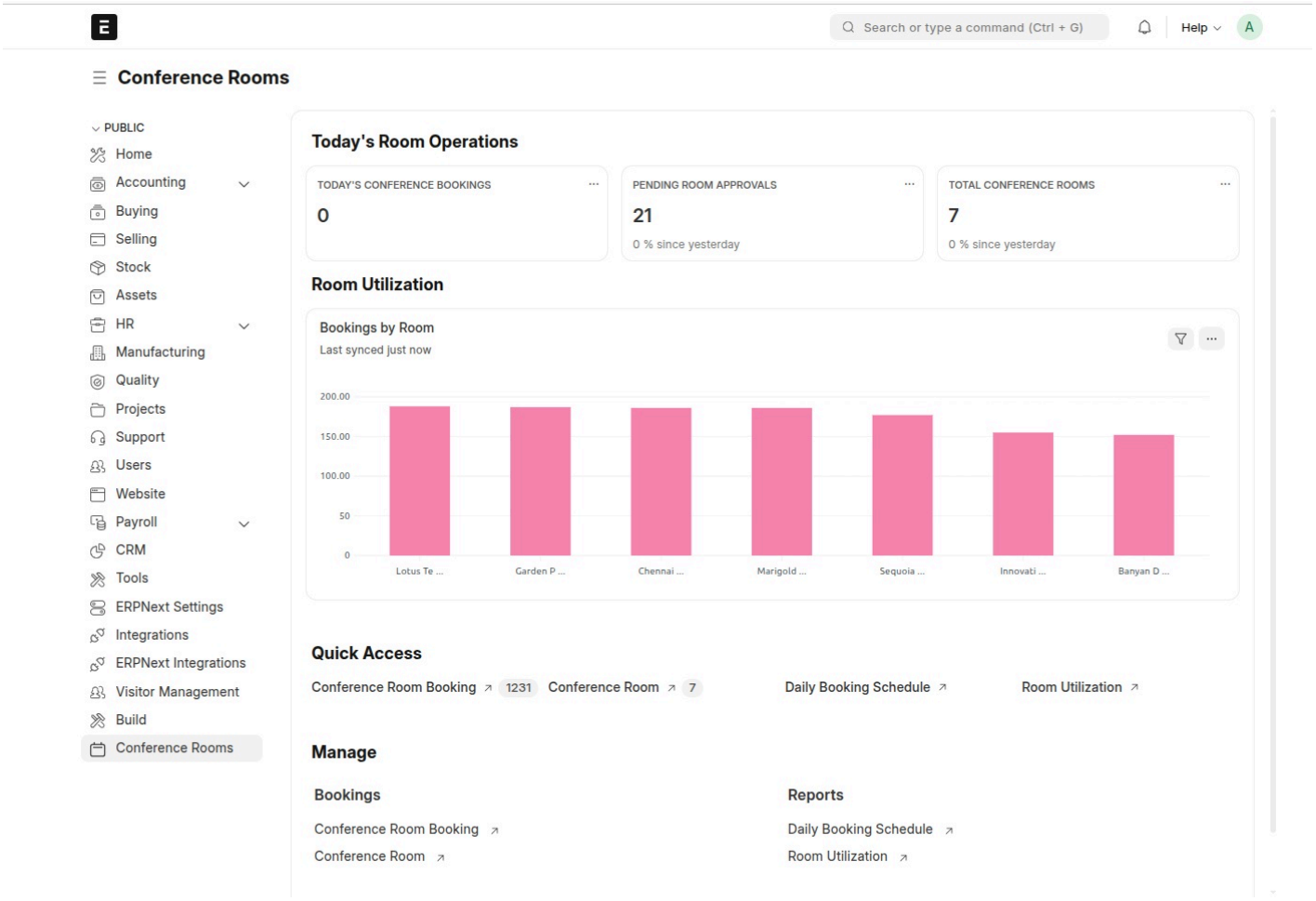


Hospitality Request — auto-created on pass approval when a hospitality flag is set

When a Visitor Pass is approved with any hospitality flag set, the system creates a Hospitality Request in `Pending Approval`.

1. Open `/app/hospitality-request`.
2. Review meal type, cab pickup/drop, hotel dates, factory tour, greeting type, etc.
3. Assign staff for execution (food dept, cab vendor, hotel partner).
4. Click **Approve** → request becomes Approved + Submitted, an email goes to the assignees.

Conference Room flow



Conference Rooms workspace — today's bookings, pending approvals, room utilization

When a Visitor Pass is approved with a Conference Room selected, the system creates a Conference Room Booking in `Pending Approval`.

1. Open the **Conference Rooms** workspace — shows today's bookings, pending approvals, total rooms, and a **Bookings by Room** chart.
2. Open the pending booking and review the time window (clamped to room operating hours) and seating capacity.
3. Click **Approve** → booking becomes Approved + Submitted.

Track every visit

Every visitor's journey is captured across three audit doctypes:

Doctype	Captures
Visitor Event Log	Every workflow + lifecycle event — pass created, approved, gate scanned, checked in, checked out, etc. Immutable.
Security Log	Every gate event — Check-In, Check-Out, Gate Transfer, Alert, Badge Collected. Photo at gate, identity match, item verification, gate name. Locked after save.
Contact Trace Record	Per visitor area-by-area movement log — visited area, time in, time out, exposure risk, close contacts. Auto-closes on Check-Out.

Click **Visitor Pass** → **Connections** on any pass to drill into all linked records.

Identity verification at the gate

The Security Log enforces three identity checks for Check-In and Check-Out:

Check	Required when
Live photo at gate	Check-In and Check-Out
Visitor matches the ID proof presented	Check-In and Check-Out
Visitor matches the photo on the approved pass	Check-In and Check-Out

Skipping any blocks the save. The pass photo, ID proof scan, and live gate photo are shown side-by-side in the **Identity Comparison** section.

Multi-Gate Routing

When a visitor arrives without a pre-assigned gate, the Security Log auto-assigns one based on visitor type:

Visitor Type	Default Gate
VIP	VIP Entrance
Supplier	Loading Dock
Contractor	Back Gate
Candidate	Main Gate
Customer	Main Gate

This can be overridden manually on each Security Log or globally via **VMS Settings** → **Default Gate**.

Visitor Blacklist

Field	What to put
Visitor Name	Person's name (used as fallback when ID number is unknown)
ID Proof Type	Aadhaar / PAN Card / Passport / Driving License
ID Proof Number	Primary lookup key (auto-named series `VB-YYYY-#####`)
Reason	Mandatory — why this person is blocked (audit/compliance requirement)
Is Active	✅ to enforce the block. Uncheck to suspend the block without deleting the record.

The blacklist is **enforced at three points**: when a Visitor Pass is submitted, when a Security Log Check-In is saved, and when a QR code is scanned at the gate.

A match by ID Proof Number blocks first; if no number is on the blacklist row, the system falls back to a case-insensitive match on Visitor Name + ID Proof Type.

Reports

Seven query reports are pre-installed:

Report	Source DocType	Use
Active Visitors	Visitor Pass	Visitors currently on premises
Daily Visitor Log	Visitor Pass	Day-by-day pass log with status
Gate Wise Count	Security Log	Check-in / check-out / inside / pending counts per gate
Visitor Identity Match Report	Visitor Pass	Audit of identity verification at the gate
Daily Hospitality Schedule	Hospitality Request	Today's cab / hotel / tour / greeting schedule
Daily Booking Schedule	Conference Room Booking	Today's room bookings
Room Utilization	Conference Room Booking	Per-room utilization summary

Limitations

- **Refunds / cancellations** of issued badges are not automated. A pass can be Cancelled via the workflow; badges are revoked on Check-Out only.
- **Recurring visitors** are tracked one Visitor Pass per visit. There is no cross-visit aggregation (e.g. "contractor X visited 12 times this year" requires a custom report on top of the Visitor Pass table).
- **NDA capture** is not implemented. A **data-retention purge** now exists but ships switched off — see Before you install; until it is enabled, visitor records remain until manually deleted.
- **Items-out reconciliation** (verifying the visitor leaves with the same items they brought in) is not enforced — only items-in is verified.
- **Currency / localisation** — the app is locale-neutral, but date formatting follows the site's Frappe locale.

Dependencies

- Frappe v16
- ERPNext v16
- HRMS v16 (Employee doctype used for visitor host links)
- Python 3.14+
- MariaDB 11.8+ with InnoDB

Support

I want to…	Go here
Report a bug	Open an issue at **https://github.com/finstein-erpnext/Finstein_Visitor_Management/issues** — include your Frappe, ERPNext, HRMS and app versions (`bench version`), the exact steps, and the traceback from **Error Log** if there is one.
Ask a question, or ask for help setting it up	Email **erpnext@finstein.ai**.
Report a security problem	Email **erpnext@finstein.ai** directly. Please do **not** open a public issue for a security report.
Read the source, or track what changed	Source: https://github.com/finstein-erpnext/Finstein_Visitor_Management · Changes: `CHANGELOG.md`

Issues are the fastest route for anything reproducible; email is better for anything that would expose your data or your site.

Third-party components

This app bundles **html5-qrcode** (Apache-2.0) for camera QR scanning at the gate, and depends on `qrcode` / `Pillow` for QR generation. Full attribution and licence text: `THIRD_PARTY_NOTICES.md`.

License

MIT — see `license.txt`.

Built with Frappe . by Finstein

Frappe Finstein